CS 310 : Automata Theory

S. Krishna

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- Assume L is regular. Then there is a DFA A with some m states $Q = \{s_1, \ldots, s_m\}$ such that $L(A) = L$, let s_1 be the initial state.

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- Then $a^j b^i$ is accepted as well.

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Pumping Lemma

Let L be a regular language. Then there is a number p called the pumping length where, if $w \in L$, $|w| \geq p$, then w can be divided into three pieces $w = xyz$ such that

(i) for all $i \geq 0$, $xy^iz \in L$,

(ii) $|y| > 0$, and

(iii) $|xy| \leq p$.

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What about finite languages?

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What if condition (ii) was not there?

Pumping lemma holds trivially

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- By pigeonhole principle, at least two states in $\underbrace{q_0, \ldots, q_p}_{p+1>n}$ should be equal.
- Therefore, there are i and j such that $0 \leq i < j \leq p$ and $q_i = q_j$.

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- Now, we break $w = xyz$ as follows.
  - $x = a_1 a_2 ... a_i$
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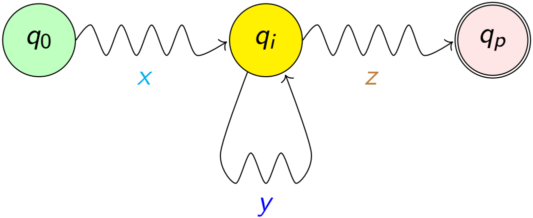

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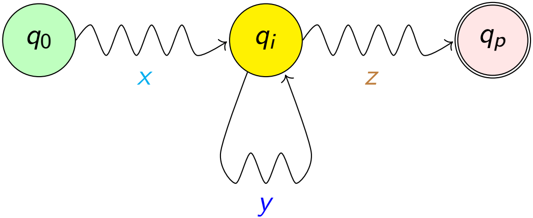

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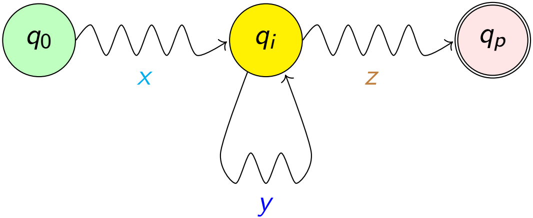

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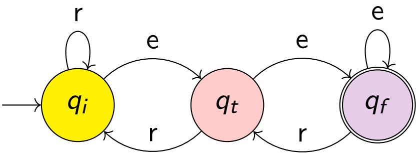
- Therefore, $xy^k z \in L$ for all $k \geq 0$.

Example: pumping words

Consider DFA



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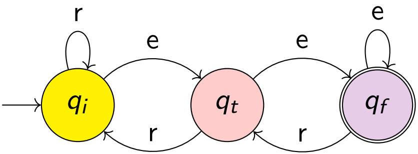

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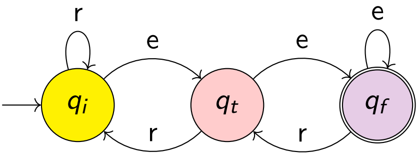

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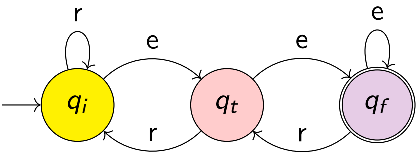

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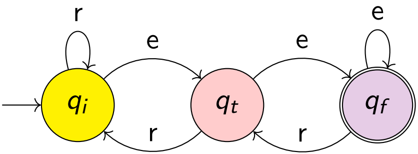

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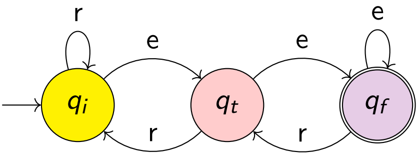

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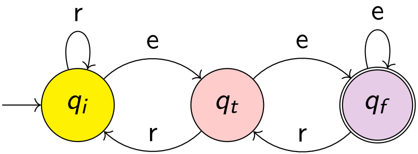

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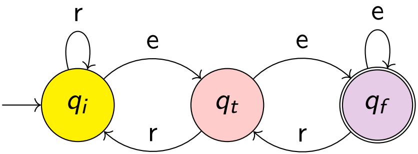

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Therefore, $e(re)^k ere \in L$.

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Its contrapositive

$$\forall p \, \exists w \in L. \big[|w| \geq p \wedge \ \forall x, y, z \big((w = xyz \wedge y \neq \epsilon \wedge |xy| \leq p) \Rightarrow \exists i \geq 0. \ xy^iz \notin L \big)\big]$$

$\Rightarrow$ L is not regular

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The instantiations are the creative steps!

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We have proven that L is not regular.

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- Clearly, $0^{p-j}1^p \notin L_{eq}$. Therefore, L_{eq} is not regular.